

AMITH M S

Backend Engineer · AI & NLP Systems · Blockchain / Web3

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PROFESSIONAL SUMMARY

Backend-focused engineer specializing in NLP-powered API systems, containerized microservice architecture, and on-chain smart contract development. Designed and shipped a production-deployed semantic resume analysis platform — a 3-service Docker system processing PDF documents in under 3 seconds using cosine-similarity embeddings and hybrid ATS scoring. Authored Sui Move smart contracts implementing provably fair on-chain randomness, full NFT lifecycle management, and a pity-guarantee mechanism backed by dynamic object storage. Participant in the **NEUROBOTS Global Robotics Championship 2026 at IIT Palakkad**. Seeking backend engineering and AI infrastructure internship roles at high-growth product companies.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), SQL, Sui Move, C#, C/C++

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, Multer, Rate Limiting, Async Processing

AI / NLP: Semantic Embeddings, Cosine Similarity, NLP Pipelines, Resume Parsing, Skill-Gap Detection, LLM Integration

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3, Chart.js, Responsive Design

Databases: MySQL, SQLite, PostgreSQL (learning), Data Modeling, Schema Design, Indexing, LocalStorage

Blockchain: Sui Move Smart Contracts, NFT Lifecycle (Mint/Transfer/Burn), On-chain Randomness, Dynamic Object Storage, Sui CLI

DevOps / Cloud: Docker, Docker Compose, Git, GitHub Actions, Vercel, Linux, Environment Parity, CI/CD

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Testing & Tools: Unit Testing, Postman, VS Code, Contract Test Suites, Edge-case Validation

EXPERIENCE

Python Engineering Intern

Jun 2025

Futura Labs · Calicut, Kerala · Short-term Internship

- Refactored 3 Python backend modules, eliminating recurring edge-case failures in data processing workflows through structured unit test coverage and systematic root-cause analysis.
- Diagnosed and resolved pipeline errors across data ingestion flows, improving code reliability and reducing manual debugging overhead during development cycles.
- Participated in code review cycles, contributing structured feedback on exception handling and input validation patterns.

PROJECTS

Deep Resume Analyzer

Python · Node.js · Express.js · FastAPI · Docker · NLP Embeddings · React

Mar–Apr 2026

Live: deep-resume-analyzer.vercel.app · GitHub: github.com/amith-m-s/deep-resume-analyzer

- Architected and deployed a production-grade semantic resume screening system using a 3-service Docker Compose stack — React frontend, Express.js API gateway, and FastAPI NLP microservice — achieving end-to-end PDF processing in under 3 seconds.
- Engineered a hybrid ATS scoring model combining cosine similarity on sentence embeddings (weight: 0.6), keyword precision matching (0.25), and domain skill scoring (0.15) to reduce false-positive candidate rankings.
- Implemented a modular NLP pipeline with text normalization, tokenization, and embedding layers — designed for stage-level extensibility and future BERT/transformer integration without rewriting core logic.

- Built skill-gap detection and structured candidate ranking output, simulating real-world recruiter pipeline evaluation across varied resume formats and job descriptions.
- Containerized the full system with Docker for environment parity across development and production; deployed on Vercel with CDN-backed static frontend delivery.

LootBox Game (On-chain)

Sui Move · Smart Contracts · NFT Architecture · On-chain Randomness · Sui CLI

Feb–Mar 2026

GitHub: github.com/amith-m-s/lootbox-game

- Designed and deployed a fully on-chain NFT loot box system on Sui blockchain using Sui Move — implementing shared GameConfig, AdminCap ownership pattern, and exact-price Coin<SUI> payment enforcement.
- Engineered provably fair randomness using sui::random with a consuming entry function architecture (non-public) to prevent front-running and ensure verifiable drop outcomes.
- Implemented a 4-tier rarity distribution (Common 60%, Rare 25%, Epic 12%, Legendary 3%) with a pity-guarantee system — ensuring Legendary drops after 30 consecutive non-Legendary opens, tracked via on-chain Table<address, u8> dynamic fields.
- Built full NFT lifecycle: mint, transfer, and burn operations with event logging; admin-controlled rarity weight updates and treasury withdrawal via AdminCap-gated functions.
- Authored comprehensive test suite covering contract initialization, invalid payment rejection, rarity boundary validation, pity threshold enforcement, and item transfer/burn correctness.

JARVIS — Modular Voice Assistant

Python · Speech Recognition · NLP · Plugin Architecture · System Automation

Jan–May 2026

GitHub: github.com/amith-m-s/JARVIS

- Built a Python-based offline voice assistant with NLP-driven intent classification, supporting system automation tasks including application launch, web search, and file operations via speech input.
- Designed a plugin-style modular command architecture — new commands extend the system without modifying core routing logic, enabling zero-downtime feature additions.

Pro Finance Tracker

HTML5 · CSS3 · Vanilla JavaScript · Chart.js · LocalStorage API

Oct–Nov 2025

GitHub: github.com/amith-m-s/Pro-Finance-Tracker

- Engineered a zero-backend personal finance dashboard with real-time Chart.js visualizations across 10+ expense categories, configurable budget thresholds, and overspend alert triggers — entirely client-side with no infrastructure cost.
- Implemented LocalStorage persistence layer with monthly trend analytics, enabling rolling spend-pattern tracking across time periods without server-side infrastructure.

HACKATHONS & EVENTS

NEUROBOTS Global Robotics Championship 2026

Jan 18, 2026

IIT Palakkad · Team: Monte Carlo · Certificate No: NEUR0101BOTS

- Represented team Monte Carlo in a global-level hackathon hosted at IIT Palakkad, working on robotics and systems engineering challenges under competitive conditions.
- Gained hands-on exposure to embedded systems design and cross-disciplinary problem-solving in a university research environment.

EDUCATION

B.Tech — Computer Science and Business Systems (CSBS)

2023 – 2027

Rajagiri School of Engineering & Technology · APJ Abdul Kalam Technological University · Kakkanad, Kerala

CGPA: 7.90 / 10.0 · Currently in Semester 7 (from Jun 2025)

Class XII — CBSE Science

2022 – 2023

Kendriya Vidyalaya No. 1, Easthill · Kozhikode, Kerala

Score: 86.40%

CERTIFICATIONS

- Microsoft Certified: Foundational C# with Microsoft — Score: 91.3%
- HackerRank Certified Software Engineer
- NPTEL: Database Management Systems — IIT Kharagpur
- Fortinet Certified Associate in Cybersecurity
- AWS Machine Learning Learning Path (in progress)

LEADERSHIP & EXTRACURRICULARS

- Volunteered in college-level events and technical activities at Rajagiri School of Engineering & Technology, contributing to student community programs and event coordination.
- Workshop attendee at IIT Palakkad (Jan 2026) — participated in technical sessions alongside the NEUROBOTS hackathon, gaining exposure to research-oriented engineering practices.